

Supplier invoice exceptions

Checking purchase-order evidence before the next handoff

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WATCH VIDEO (4:24)

<https://drive.google.com/file/d/1zqXr-ju4dSJ-dirIIUqGpt7z8tIBcYBC/view>

The documented handoff

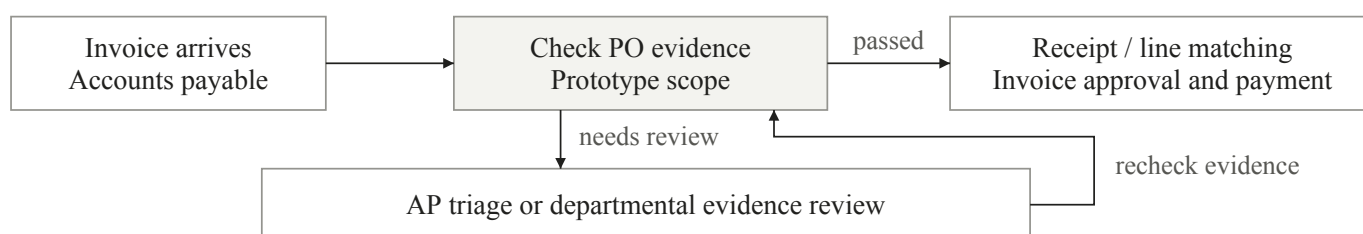
Buckinghamshire Healthcare NHS Trust (BHT) reports **275 purchase-order (PO) lines raised after invoice date** in June 2026, versus 214 in May. Its accounts payable (AP) staff follow up with departments for PO creation; finding the right department can delay payment. [1] These are line counts, not an invoice exception rate.

Why start here

I chose a decision that can be checked case by case: **which evidence needs review, and by whom.** Prevention may matter more. April's report attributes rising unapproved invoices mainly to delayed annual PO setup. [2] I would compare this queue with any existing AP report on the same cases, measuring decision time and routing errors. If that report supports the same decisions, or another handoff takes more staff time, I would change priorities.

Reference workflow

Purchase request → authorised PO → supply / receipt



Simplified from public descriptions [1, 3]. The Trust's exact systems and handoffs are unverified.

Who would use this

Assumed roles and goals. No stakeholder interviews have taken place.

Role	Goal and constraint
AP manager · proposed owner	Prioritise PO problems seen at invoice arrival within available staffing.
AP processor	Review evidence and identify the responsible role. Cannot authorise a purchase.
Department buyer / budget holder	Confirm the reference or approval evidence within delegated authority.
Supplier · affected party	Predictable payment. Cannot repair the buyer's internal records.

Scope and success

I check PO references, supplier and purchase-request IDs, currency and approval evidence for routine GBP invoices with positive amounts. Capital, emergency, PO-exempt and credit documents are excluded. Unused PO value, line matching, receipts, disputes, tax and payment are outside scope.

Still unknown: real workload, existing tools, process owner and added value. Operational data and policy rules are synthetic; BHT is not a client.

Diagnostic KPI: confirmed PO exceptions at invoice arrival, divided by eligible arrivals with enough evidence to assess. UNKNOWN is excluded from this denominator.

For open exceptions, I report reason counts, gross value, median age since invoice arrival and the share with a known department reviewer role. I also report assessment coverage across all eligible arrivals.

Success: correct, traceable outcomes for every named case and identical model records and metrics on rerun.

[1] BHT, June 2026 finance report, pp. 15-16.

[2] BHT, April 2026 finance report, p. 15.

[3] HMRC, *Procure to pay*, GfC8 part 4. Research checked 18 September 2026.

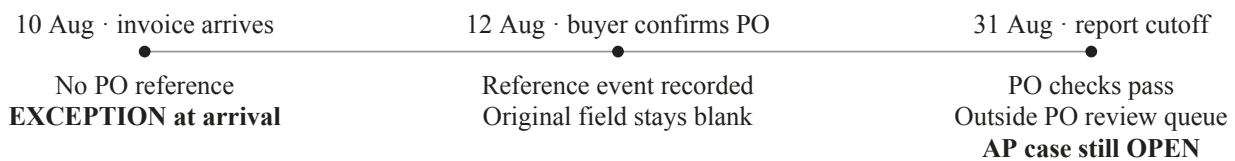
The data workflow

I generated 100 synthetic invoices, including 40 named edge cases. An exact duplicate and a conflicting copy bring the raw export to 102 rows.

Source of truth in the simulation	Record structure and key fields
Procurement · SQLite via SQL	PO/request/department links, supplier, currency and creation time. State events have effective and recorded times.
AP · invoice CSV	Invoice, supplier and original PO/request IDs; issue/arrival times, amount in integer minor units and currency.
AP · case-event CSV	Invoice and event IDs, actor, event type, effective time and recorded time.

SQL + CSV → preserve raw files → validate → model → metrics and review

Example: a reference confirmed after arrival (not to scale)



I summarise histories before joining records to avoid double-counting. Each invoice has one row with arrival and cutoff assessments. AP closure needs its own event; receipt, allocation and payment events are absent.

Handling incomplete records

Whitespace, identifier case and timezone fixes are logged. Exact duplicates are removed; malformed or conflicting rows are quarantined. Gaps or contradictions in decision history can produce UNKNOWN. An approval entered after an invoice arrives is not treated as evidence available when it arrived.

From **102 raw rows**, I retained **97**, removed one exact duplicate and quarantined four. Seven exclusions leave **90 eligible invoices**.

Reliability and reproduction

52 tests pass, covering 40 separately specified scenarios. Reruns produce identical model, metric and review files. On an input hash mismatch, the pipeline records FAILED and retains the rejected input and last successful output. Restoring input reproduces it.

Run `python3 run.py run`,
then `python3 run.py verify`.

Code, data and run evidence

github.com/mohit-1710/fde-invoice-evidence

Python 3.9+, offline; start with the README.

[Watch narrated demo \(4:24\)](#)

What the synthetic run shows

Arrival cohort: 1 August 2026 00:00 to 31 August 2026 18:00 UTC, inclusive. Review at the cutoff covers this cohort, not the whole AP backlog.

Measure	Result and interpretation
KPI · arrival exception rate	37 / 79 = 46.84% . The denominator excludes 11 UNKNOWN records.
Arrival assessment coverage	79 / 90 = 87.78% . UNKNOWN stays in this denominator.
Current open exceptions by reason	28 : reference missing 9, PO unapproved 7, PO cancelled 7, other 5.
Their gross invoice value	£42,454.05 . This is not unpaid value or loss.
Median age since invoice arrival	21.33 days for the 28 open exceptions.
Department review role identified	23 / 28 = 82.14% . AP must identify the role for five cases.

The cutoff queue has **28 open confirmed exceptions and 9 uncertain cases**. Evidence and case states change after arrival. The repository's methods reconcile the counts against the invoice trace.